IGCSE 0607 Practice Questions — E4.6 — Combined Questions & Markscheme

Perpendicular lines

Total Questions: 18 — **Total Marks:** 114 Marks

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  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

1 Line L has the equation $5y + 2x = 15$.
Find the gradient of a line that is perpendicular to L.

..... **[2]**

2 Find the equation of the line that is perpendicular to $y = 3x - 5$ and passes through the point $(6, 2)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

3 Line L_1 has the equation $y = 5 - 2x$.
Line L_2 is perpendicular to L_1 and passes through the point $(4, 1)$.

(a) Find the gradient of line L_1.

..... **[1]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

4 Line L_1 has the equation $4y = 3x - 8$.

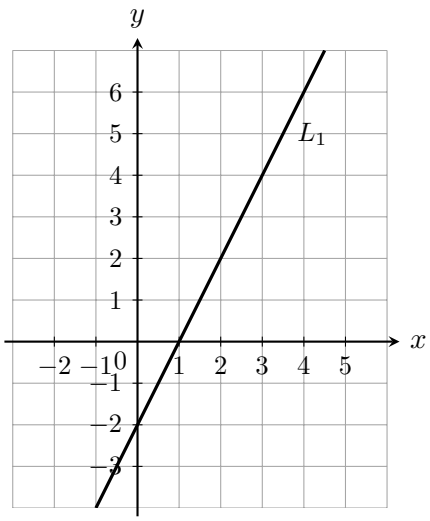
(a) Find the gradient of a line that is perpendicular to L_1.

Gradient = ... **[2]**

(b) Line L_2 is perpendicular to L_1 and passes through the point $(6, -5)$.
Find the equation of line L_2 in the form $y = mx + c$.

$y =$ ..... **[2]**

5 The diagram shows a straight line L_1 drawn on a coordinate grid.



(a) Write down the gradient of line L_1.

..... **[1]**

(b) Find the gradient of a line that is perpendicular to L_1.

..... **[1]**

(c) Line L_2 is perpendicular to L_1 and passes through the point $(4, 3)$.
Find the equation of line L_2 in the form $y = mx + c$.

$y =$ ..... **[3]**

6 A straight line L has the equation $2y = 5x + 8$.

(a) Find the gradient of line L.

..... **[2]**

(b) Find the gradient of a line that is perpendicular to line L.

..... **[1]**

(c) A line M is perpendicular to line L and passes through the point $(10, -3)$.
Find the equation of line M in the form $y = mx + c$.

$y =$ ..... **[3]**

7 Point A has coordinates $(-3, 8)$ and point B has coordinates $(9, -2)$.

(a) Find the coordinates of the midpoint of AB and the gradient of AB.

Midpoint = ... , Gradient = ... **[2]**

(b) Find the equation of the perpendicular bisector of AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[3]**

8 A is the point $(-4, 7)$ and B is the point $(2, -1)$.

Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[5]**

9 Line L has equation $2x + 3y = 12$.
Point A has coordinates $(-3, 8)$ and point B has coordinates $(9, -2)$.

(a) Find the gradient of a line that is perpendicular to line L.

..... **[2]**

(b) For the line segment joining $A(-3, 8)$ and $B(9, -2)$,

(i) find the coordinates of the midpoint of AB,

(..... , ...) **[2]**

(ii) find the equation of the perpendicular bisector of AB, giving your answer in the form $ax + by = c$, where a, b, and c are integers.

..... **[4]**

10 A is the point $(-3, 8)$ and B is the point $(9, -2)$.

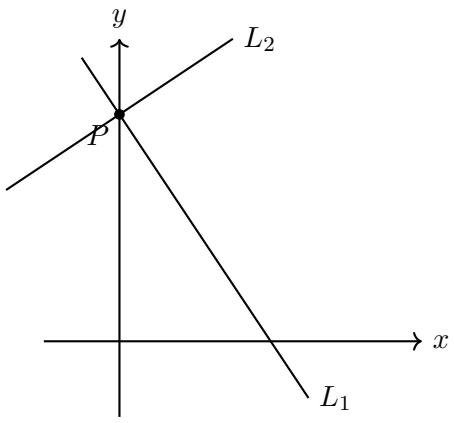
(a) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

(b) Find the equation of the perpendicular bisector of AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[4]**

11 Line L_1 has the equation $3x + 2y = 12$.
Line L_2 is perpendicular to L_1 and intersects L_1 at a point P on the y-axis.



Not to scale

(a) Find the coordinates of point P.

(..... , ...) **[1]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Point Q lies on L_2 and has an x-coordinate of 9.
Point R lies on L_1 and has an x-coordinate of 4.
Work out the area of triangle PQR.

..... **[5]**

12 Line L_1 has equation $kx + 2y = 12$.
Line L_2 passes through the points $A(0, 5)$ and $B(2, k + 2)$.
The constant k is a non-zero integer.

(a) Find the gradient of line L_1 in terms of k.

..... **[1]**

(b) Find the gradient of line L_2 in terms of k.

..... **[2]**

(c) Given that L_1 and L_2 are perpendicular, find the two possible values of k.

$k =$ ..... **[4]**

(d) Given that the gradient of line L_1 is positive, find the equation of line L_2 in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

13 The equation of line P is $3x - 4y = 12$.
Line Q is perpendicular to line P and passes through the point $(6, -2)$.

Find the equation of line Q.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

14 The points A and B have coordinates $(-2, 4)$ and $(6, 0)$ respectively.

(a) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

(b) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

15 The coordinates of point A are $(-4, 5)$ and the coordinates of point B are $(2, -7)$.

(a) Find the coordinates of the midpoint of the line segment AB.

(..... , ...) **[2]**

(b) Calculate the gradient of the line AB.

..... **[2]**

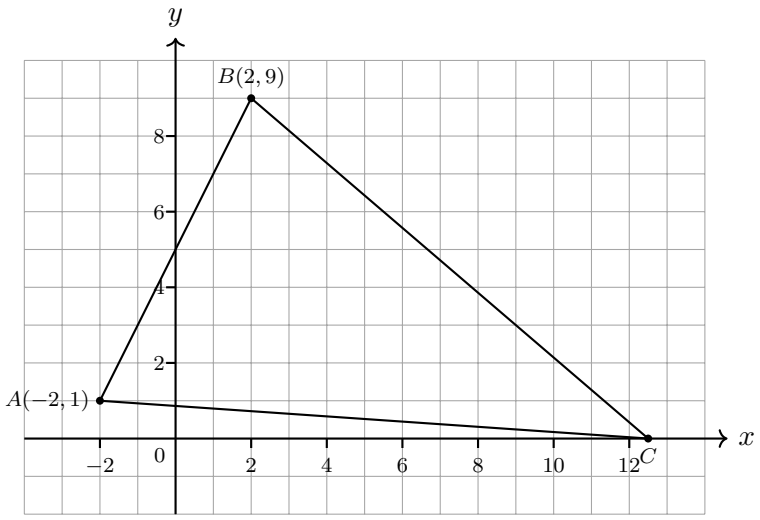
(c) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(d) The perpendicular bisector intersects the x-axis at point P.
Find the coordinates of P.

(..... , ...) **[2]**

16 The diagram shows a right-angled triangle ABC with $\angle ABC = 90°$.
The coordinates of A and B are $(-2, 1)$ and $(2, 9)$ respectively, and point C lies on the x-axis.



(a) Find the equation of the line BC.

$y =$ ..... **[3]**

(b) Find the coordinates of point C.

(..... , ...) **[2]**

(c) Calculate the area of triangle ABC.

Area = ... **[2]**

17 The line L_1 has equation $2y + 3x = 12$.
The line L_2 is perpendicular to L_1 and intersects L_1 at the point P on the y-axis.

(a) Find the coordinates of P.

(..... , ...) **[2]**

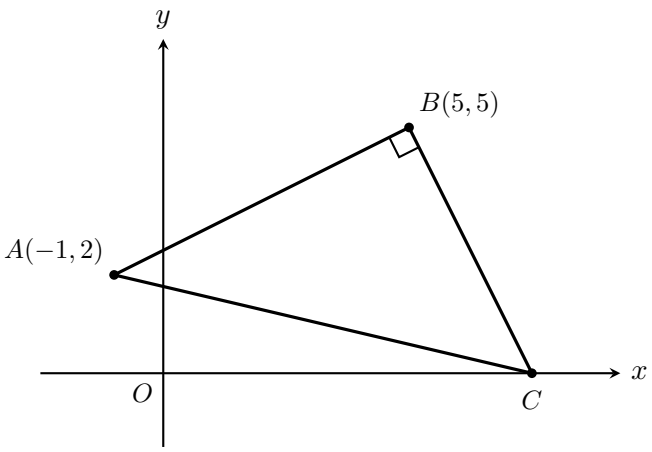
(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Line L_2 crosses the x-axis at the point Q.
Calculate the exact length of PQ.

..... **[4]**

18 The diagram shows a sketch of triangle ABC with vertices $A(-1, 2)$ and $B(5, 5)$.
Angle $ABC = 90°$ and vertex C lies on the x-axis.



(a) Find the equation of the line passing through B and C in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Find the coordinates of vertex C.

$C =$ (..... , ...) **[2]**

(c) Calculate the area of triangle ABC.

..... square units **[3]**

(d) The perpendicular bisector of AC intersects the y-axis at point P.

Find the coordinates of P, giving the y-coordinate correct to 3 significant figures.

$P = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[4]**

1 Rearranges to $5y = -2x + 15$ to find gradient of L is $-\frac{2}{5}$ **[M1]**

$\frac{5}{2}$ or 2.5 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 Identifies the gradient of the given line as 3 **[B1]**

Uses the perpendicular gradient $-\frac{1}{3}$ and substitutes $(6, 2)$ into $y = mx + c$ or $y - y_1 = m(x - x_1)$ **[M1]**

$-\frac{1}{3}x + 4$ **[A1]**

[Back to Question] *[Total Marks: 3]*

3 (a) -2 **[B1]**

(b) Gradient of $L_2 = \frac{1}{2}$ **[M1]**

$1 = \frac{1}{2}(4) + c$ or $y - 1 = \frac{1}{2}(x - 4)$ **[M1]**

$y = \frac{1}{2}x - 1$ **[A1]**

[Back to Question] *[Total Marks: 4]*

4 (a) Finds gradient of L_1: $y = \frac{3}{4}x - 2$ giving $m_1 = \frac{3}{4}$ **[M1]**

$-\frac{4}{3}$ **[A1]**

(b) Substitutes $(6, -5)$ and $m = -\frac{4}{3}$ into $y = mx + c$: $-5 = -\frac{4}{3}(6) + c$ **[M1]**

$-\frac{4}{3}x + 3$ **[A1]**

[Back to Question] *[Total Marks: 4]*

5 (a) 2 **[B1]**

(b) $-\frac{1}{2}$ (or -0.5) **[B1]**

(c) $3 = -\frac{1}{2}(4) + c$ or $y - 3 = -\frac{1}{2}(x - 4)$ **[M1]**

$c = 5$ **[A1]**

$-\frac{1}{2}x + 5$ **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** Rearranges the equation to make y the subject: $y = \frac{5}{2}x + 4$ **[M1]**

$\frac{5}{2}$ or 2.5 **[A1]**

(b) $-\frac{2}{5}$ or -0.4 **[B1]**

(c) Substitutes $(10, -3)$ and $m = -\frac{2}{5}$ into linear equation: $-3 = -\frac{2}{5}(10) + c$ or $y - (-3) = -\frac{2}{5}(x - 10)$ **[M1]**

Solves for c: $-3 = -4 + c \implies c = 1$ **[M1]**

$-\frac{2}{5}x + 1$ (or $-0.4x + 1$) **[A1]**

[Back to Question] *[Total Marks: 6]*

7 **(a)** Midpoint $= \left(\frac{-3+9}{2}, \frac{8+(-2)}{2}\right) = (3, 3)$ **[B1]**

Gradient $= \frac{-2-8}{9-(-3)} = -\frac{10}{12} = -\frac{5}{6}$ **[B1]**

(b) Uses perpendicular gradient $m_\perp = \frac{6}{5}$ **[M1]**

Substitutes midpoint $(3, 3)$: $3 = \frac{6}{5}(3) + c$ or $y - 3 = \frac{6}{5}(x - 3)$ **[M1]**

$6x - 5y = 3$ (or $-6x + 5y = -3$) **[A1]**

[Back to Question] *[Total Marks: 5]*

8 Calculates the midpoint of $AB = \left(\frac{-4+2}{2}, \frac{7+(-1)}{2}\right) = (-1, 3)$ **[B1]**

Calculates the gradient of $AB = \frac{-1-7}{2-(-4)} = -\frac{8}{6} = -\frac{4}{3}$ **[B1]**

Finds the perpendicular gradient $= \frac{3}{4}$ **[M1]**

Substitutes the midpoint $(-1, 3)$ and gradient $\frac{3}{4}$ to find the equation, e.g., $y - 3 = \frac{3}{4}(x - (-1))$ **[M1]**

$3x - 4y = -15$ or $4y - 3x = 15$ or equivalent integer form **[A1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** Gradient of $L = -\frac{2}{3}$ **[M1]**

$\frac{3}{2}$ (or 1.5) **[A1]**

(b) **(i)** $\left(\frac{-3+9}{2}, \frac{8+(-2)}{2}\right)$ **[M1]**

$(3, 3)$ **[A1]**

(ii) Gradient of $AB = \dfrac{-2-8}{9-(-3)} = -\dfrac{5}{6}$ **[M1]**

Gradient of perpendicular bisector $= \dfrac{6}{5}$ **[A1]**

$y - 3 = \dfrac{6}{5}(x-3)$ **[M1]**

$6x - 5y = 3$ (or $-6x + 5y = -3$) **[A1]**

[Back to Question] *[Total Marks: 8]*

10 **(a)** $\left(\dfrac{-3+9}{2}, \dfrac{8+(-2)}{2}\right)$ **[M1]**

$(3, 3)$ **[A1]**

(b) Gradient of $AB = \dfrac{-2-8}{9-(-3)}$ or $-\dfrac{10}{12}$ or $-\dfrac{5}{6}$ **[M1]**

Perpendicular gradient is $\dfrac{6}{5}$ **[M1]**

Substitute $(3, 3)$ into $y - 3 = \dfrac{6}{5}(x-3)$ or $3 = \dfrac{6}{5}(3) + c$ **[M1]**

$6x - 5y = 3$ or $-6x + 5y = -3$ **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** $(0, 6)$ **[B1]**

(b) Gradient of $L_1 = -\dfrac{3}{2}$ **[M1]**

Gradient of $L_2 = \dfrac{2}{3}$ **[M1]**

$y = \dfrac{2}{3}x + 6$ **[A1]**

(c) Coordinates of Q: $y = \dfrac{2}{3}(9) + 6 = 12 \implies Q(9, 12)$ **[B1]**

Coordinates of R: $3(4) + 2y = 12 \implies y = 0 \implies R(4, 0)$ **[B1]**

Identify triangle PQR is right-angled at P OR setup valid box/shoelace method **[M1]**

Length $PQ = \sqrt{9^2 + 6^2} = \sqrt{117}$ and Length $PR = \sqrt{4^2 + (-6)^2} = \sqrt{52}$ **[M1]**

Area $= \dfrac{1}{2} \times \sqrt{117} \times \sqrt{52} = 39$ **[A1]**

[Back to Question] *[Total Marks: 9]*

12 **(a)** Rearranges to $2y = -kx + 12$ to identify gradient $-\dfrac{k}{2}$ **[B1]**

(b) Uses the gradient formula: $\dfrac{(k+2)-5}{2-0}$ **[M1]**

$\dfrac{k-3}{2}$ **[A1]**

(c) Applies the perpendicularity condition $m_1 \times m_2 = -1$ **[M1]**

Forms the equation: $\left(-\dfrac{k}{2}\right)\left(\dfrac{k-3}{2}\right) = -1$ **[M1]**

Expands and rearranges to form a quadratic equation:
$-k(k-3) = -4 \implies k^2 - 3k - 4 = 0$ **[M1]**

4 and -1 **[A1]**

(d) Deduces that $k = -1$ must be chosen because $m_1 = -\dfrac{k}{2}$ must be positive (yielding $m_1 = \dfrac{1}{2}$) **[M1]**

Calculates the gradient of L_2 when $k = -1$: $m_2 = \dfrac{-1-3}{2} = -2$, and uses the y-intercept from point $A(0,5)$ to write $y = -2x + 5$ **[M1]**

$2x + y = 5$ **[A1]**

[Back to Question] *[Total Marks: 10]*

13 $4y = 3x - 12$ implies gradient of P is $\dfrac{3}{4}$ **[M1]**

Gradient of Q is $-\dfrac{4}{3}$ **[M1]**

Substitute $(6,-2)$ into $y = -\dfrac{4}{3}x + c$ or $y - (-2) = -\dfrac{4}{3}(x-6)$ **[M1]**

$-\dfrac{4}{3}x + 6$ **[A1]**

[Back to Question] *[Total Marks: 4]*

14 (a) $\left(\dfrac{-2+6}{2}, \dfrac{4+0}{2}\right)$ **[M1]**

$(2,2)$ **[A1]**

(b) Gradient of $AB = \dfrac{0-4}{6-(-2)} = -\dfrac{1}{2}$ **[M1]**

Gradient of perpendicular bisector $= 2$ **[M1]**

Substitute midpoint $(2,2)$ into $y - y_1 = m(x - x_1)$: $y - 2 = 2(x-2)$ **[M1]**

$y = 2x - 2$ **[A1]**

[Back to Question] *[Total Marks: 6]*

15 (a) Uses the midpoint formula: $\left(\dfrac{-4+2}{2}, \dfrac{5+(-7)}{2}\right)$ **[M1]**

$(-1,-1)$ **[A1]**

(b) Uses the gradient formula: $\dfrac{-7-5}{2-(-4)}$ or $\dfrac{5-(-7)}{-4-2}$ **[M1]**

$\frac{-12}{6} = -2$ **[A1]**

(c) Identifies the perpendicular gradient is $\frac{1}{2}$ (or 0.5) **[B1]**

Substitutes their midpoint $(-1, -1)$ and perpendicular gradient into $y = mx + c$:
$-1 = \frac{1}{2}(-1) + c$ **[M1]**

$\frac{1}{2}x - \frac{1}{2}$ (or $0.5x - 0.5$) **[A1]**

(d) Sets $y = 0$ in their equation from part (c): $0 = \frac{1}{2}x - \frac{1}{2}$ **[M1]**

$(1, 0)$ **[A1]**

[Back to Question] *[Total Marks: 9]*

16 (a) Calculates gradient of AB: $\frac{9-1}{2-(-2)} = 2$ **[B1]**

Uses perpendicular gradient $m_{BC} = -\frac{1}{2}$ (or -0.5) **[M1]**

Substitutes $B(2, 9)$: $9 = -\frac{1}{2}(2) + c$, giving $y = -\frac{1}{2}x + 10$ (or $-0.5x + 10$) **[A1]**

(b) Substitutes $y = 0$ into equation of BC: $0 = -0.5x + 10$ **[M1]**

$(20, 0)$ **[A1]**

(c) Finds $AB = \sqrt{4^2 + 8^2} = \sqrt{80}$ and $BC = \sqrt{18^2 + (-9)^2} = \sqrt{405}$ **[M1]**

$\frac{1}{2} \times \sqrt{80} \times \sqrt{405} = 90$ **[A1]**

[Back to Question] *[Total Marks: 7]*

17 (a) Substitutes $x = 0$ into $2y + 3x = 12$ because P is on the y-axis **[M1]**

$(0, 6)$ **[A1]**

(b) Rearranges L_1 equation to find gradient: $2y = -3x + 12 \implies y = -\frac{3}{2}x + 6$, so $m = -\frac{3}{2}$ **[B1]**

States perpendicular gradient for L_2 is $\frac{2}{3}$ **[M1]**

$\frac{2}{3}x + 6$ **[A1]**

(c) Substitutes $y = 0$ into the equation of L_2 to find the coordinates of Q **[M1]**

$0 = \frac{2}{3}x + 6 \implies x = -9$ **[A1]**

Uses the distance formula for $P(0, 6)$ and $Q(-9, 0)$: $\sqrt{(-9-0)^2 + (0-6)^2}$ **[M1]**

$\sqrt{117}$ or $3\sqrt{13}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

18 **(a)** Gradient of $AB = \dfrac{5-2}{5-(-1)} = \dfrac{1}{2}$ **[M1]**

Gradient of $BC = -2$ **[A1]**

$-2x + 15$ **[A1]**

(b) $0 = -2x + 15$ **[M1]**

$(7.5, 0)$ **[A1]**

(c) $AB = \sqrt{(5-(-1))^2 + (5-2)^2} = \sqrt{45}$ and $BC = \sqrt{(7.5-5)^2 + (0-5)^2} = \sqrt{31.25}$ **[M1]**

Area $= \dfrac{1}{2} \times \sqrt{45} \times \sqrt{31.25}$ **[M1]**

18.75 **[A1]**

(d) Midpoint of $AC = \left(\dfrac{-1+7.5}{2}, \dfrac{2+0}{2}\right) = (3.25, 1)$ **[B1]**

Gradient of $AC = \dfrac{0-2}{7.5-(-1)} = -\dfrac{4}{17}$, so perpendicular gradient $= \dfrac{17}{4}$ $(= 4.25)$ **[M1]**

$y - 1 = 4.25(0 - 3.25)$ **[M1]**

$(0, -12.8)$ **[A1]**

[Back to Question] *[Total Marks: 12]*